



Topic 5 – Cloud Infrastruc ture Management

Managing the Invisible
Backbone of the Cloud

Introduction to Cloud Infrastructure

Cloud infrastructure includes:

- Compute (VMs, containers)
- Storage (block, object, file)
- Network (virtual switches, routers)
- Managed via web portals or APIs

Think of it like renting a fully furnished apartment. You don't own the building (hardware), but you get everything you need inside (VM, storage, etc.)



Why Cloud Infrastructure Management Matters

- Ensures resources are:
 - Available
 - Secure
 - Efficient
- Prevents downtime and optimizes cost
- **Real-life example:**
Netflix uses cloud infrastructure to stream to millions. Without solid management, you'd be watching a spinning buffer wheel all day.



What is a Virtual Infrastructure Manager (VIM) ?

- Software that manages virtualized resources in a cloud

Examples: OpenStack, VMware vSphere, Proxmox, Eucalyptus

Acts as the “invisible caretaker” of virtual resources

Core Features of a VIM



Provision
ing VMs



Networkin
g
configura
tion



Storage
managemen
t



Resource
schedulin
g



Monitorin
g and
logging



APIs for
automatio
n

Example – OpenStack as a VIM



Open-source VIM



Modular: Nova (compute),
Neutron (network),
Cinder (storage)



Scalable and
customizable

VMware vSphere as a VIM



ENTERPRISE-GRADE



INTUITIVE GUI



INTEGRATION WITH OTHER
VMWARE TOOLS (VCENTER,
VSAN)

Feature Comparison of VIMs

Feature	OpenStack	VMware vSphere	Proxmox
Open-source	✓	✗	✓
GUI	Moderate	Excellent	Good
Enterprise support	Moderate	Excellent	Moderate
Scalability	High	High	Medium

Advantages of Using Virtual Machines



ISOLATION
BETWEEN
WORKLOADS



BETTER
UTILIZATION OF
HARDWARE



SNAPSHOT &
ROLLBACK



DISASTER
RECOVERY



PORTABILITY

Purpose of Virtual Machines



RUN MULTIPLE OS ON
ONE PHYSICAL
MACHINE



SIMULATE DIFFERENT
ENVIRONMENTS



COST-EFFICIENT
DEPLOYMENT

Benefits of Virtual Machines in Cloud



Scalability: Clone in minutes



Load balancing:
Migrate VMs based on demand



Maintenance: Update without downtime

Opportunities in Cloud Infrastructure Management



Automated
scaling



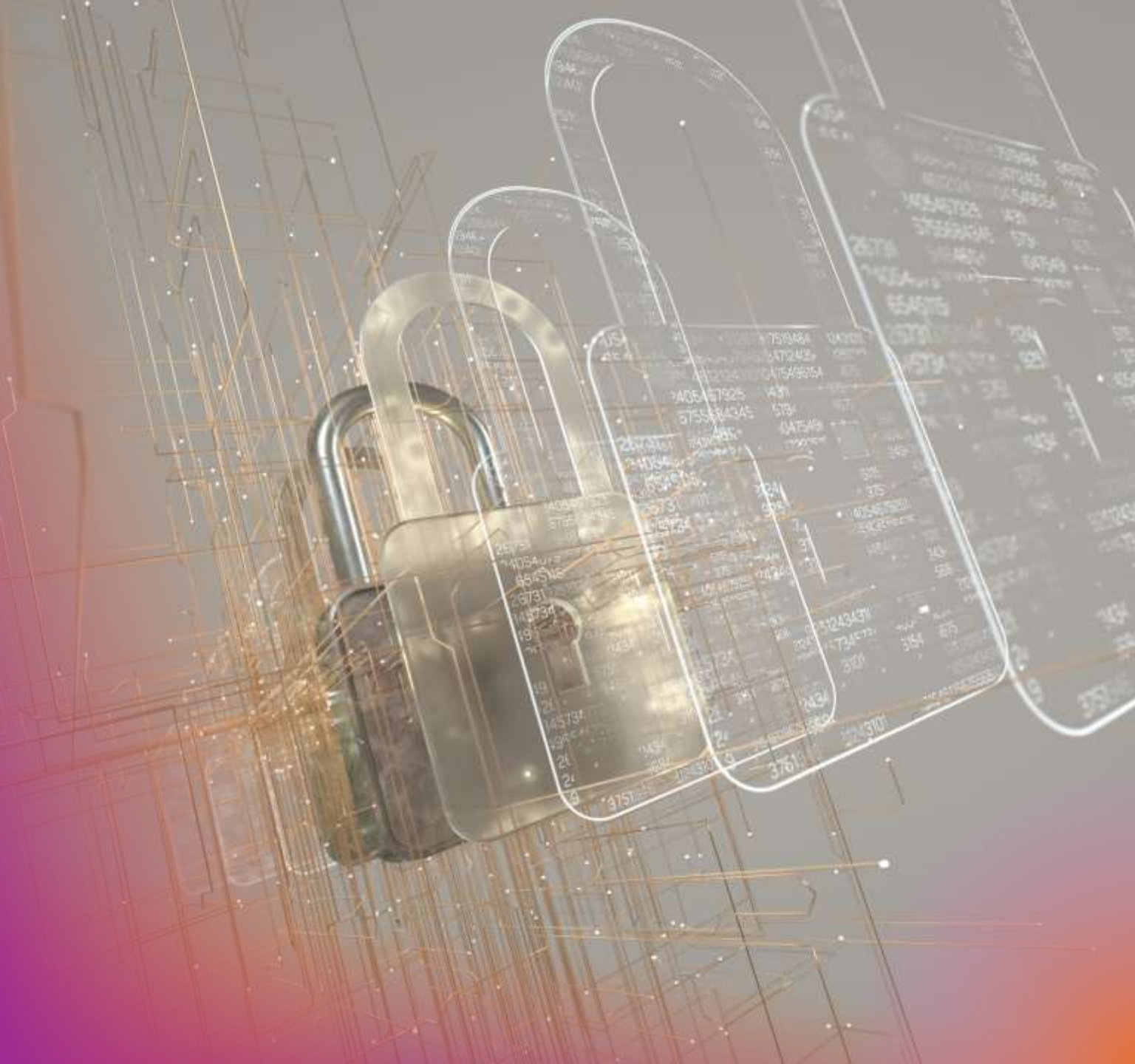
Multi-cloud
strategy



AI-based
resource
optimization



Integration
with DevOps
(CI/CD)



Challenges in Cloud Infrastructur e

- Vendor lock-in
- Complex configurations
- Latency issues
- Skills gap

Security Issues in Virtual Infrastruct ure



Hypervisor attacks



VM escape



Insecure APIs



Data breaches



Performance Issues

- Overprovisioning
- VM sprawl (too many unused VMs)
- Resource contention



Cost Management Challenges

- Pay-per-use can spiral
- Unused VMs add cost
- Lack of visibility
= billing shocks

Disadvantages of Virtual Machines

- Slower than containers
- Resource overhead
- Complex network setups
- Longer boot times



How to Overcome the Challenges



Use automation
tools (Ansible,
Terraform)



Monitor usage
and set alerts



Train staff
continuously



Use hybrid
cloud wisely

Cloud Management Best Practices

- Document configurations
- Use tagging and naming conventions
- Regular cost reviews
- Disaster recovery planning



Summary

- Cloud infrastructure = virtualized compute, storage, and network
- VIMs help manage resources
- VMs offer isolation and flexibility
- But also come with costs and complexity





Final Thoughts

“Managing cloud is like playing chess blindfolded...
unless you have the right tools!”

Explore hands-on tools like:

- VirtualBox
 - Proxmox
 - OpenStack DevStack
 - AWS Free Tier
-